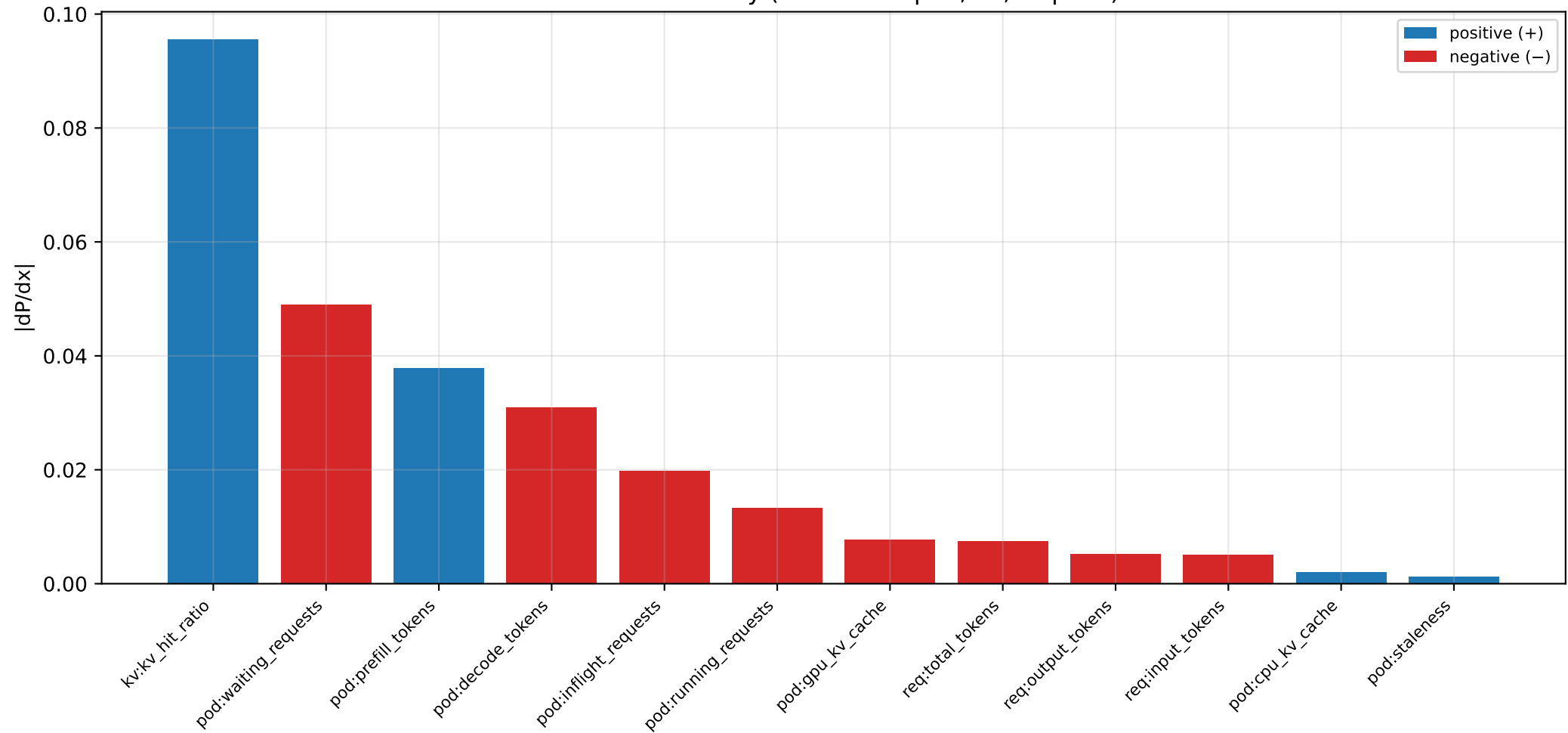


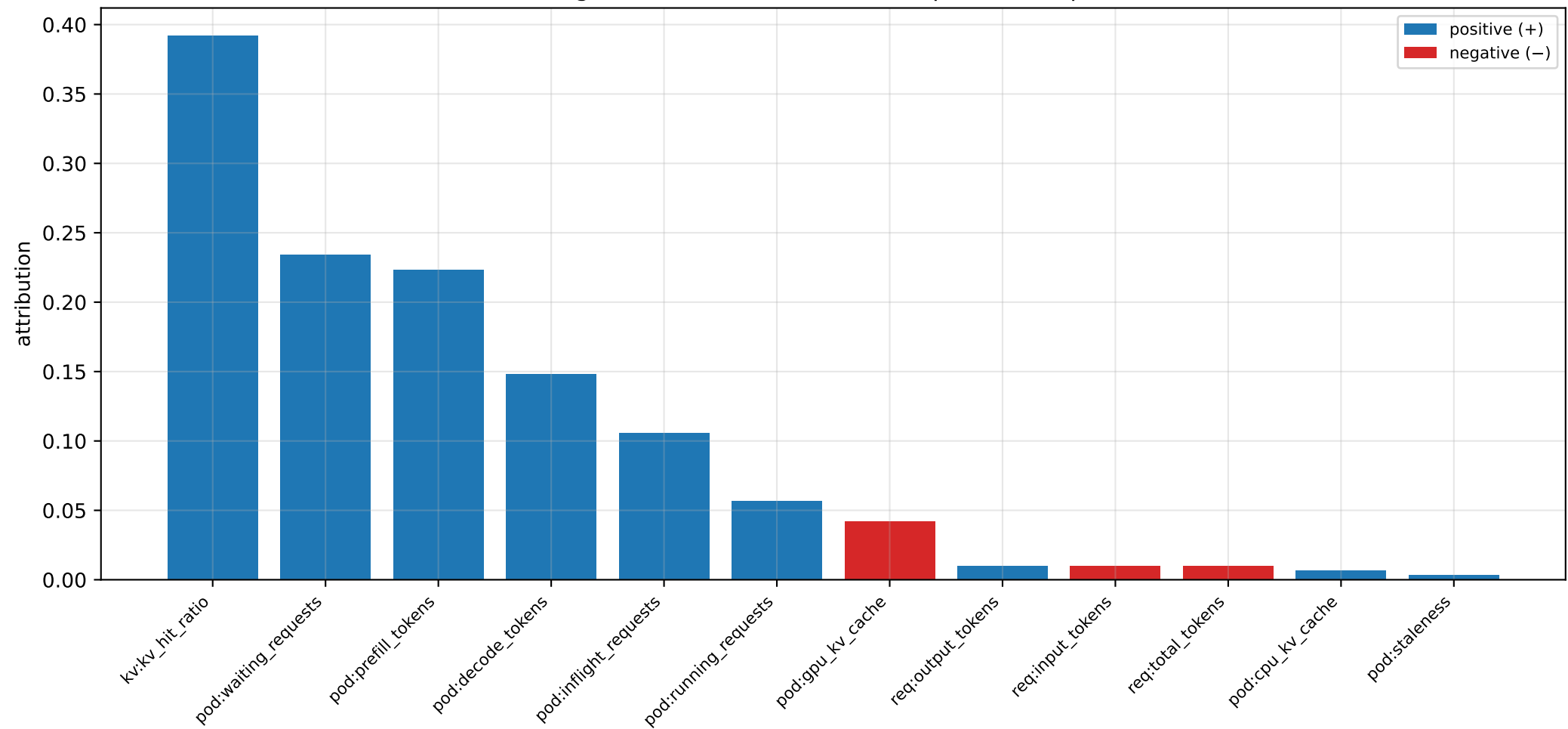
Routing Policy Explainability Report

final_model-data_replaced-processed-linear_simple-lr_0.001-ttft_weight_2.0-ttftslo_1000-avgtpotslo_50-without_none

Gradient Saliency (combined: pod, kv, request)



Integrated Gradients (combined: pod, kv, request)



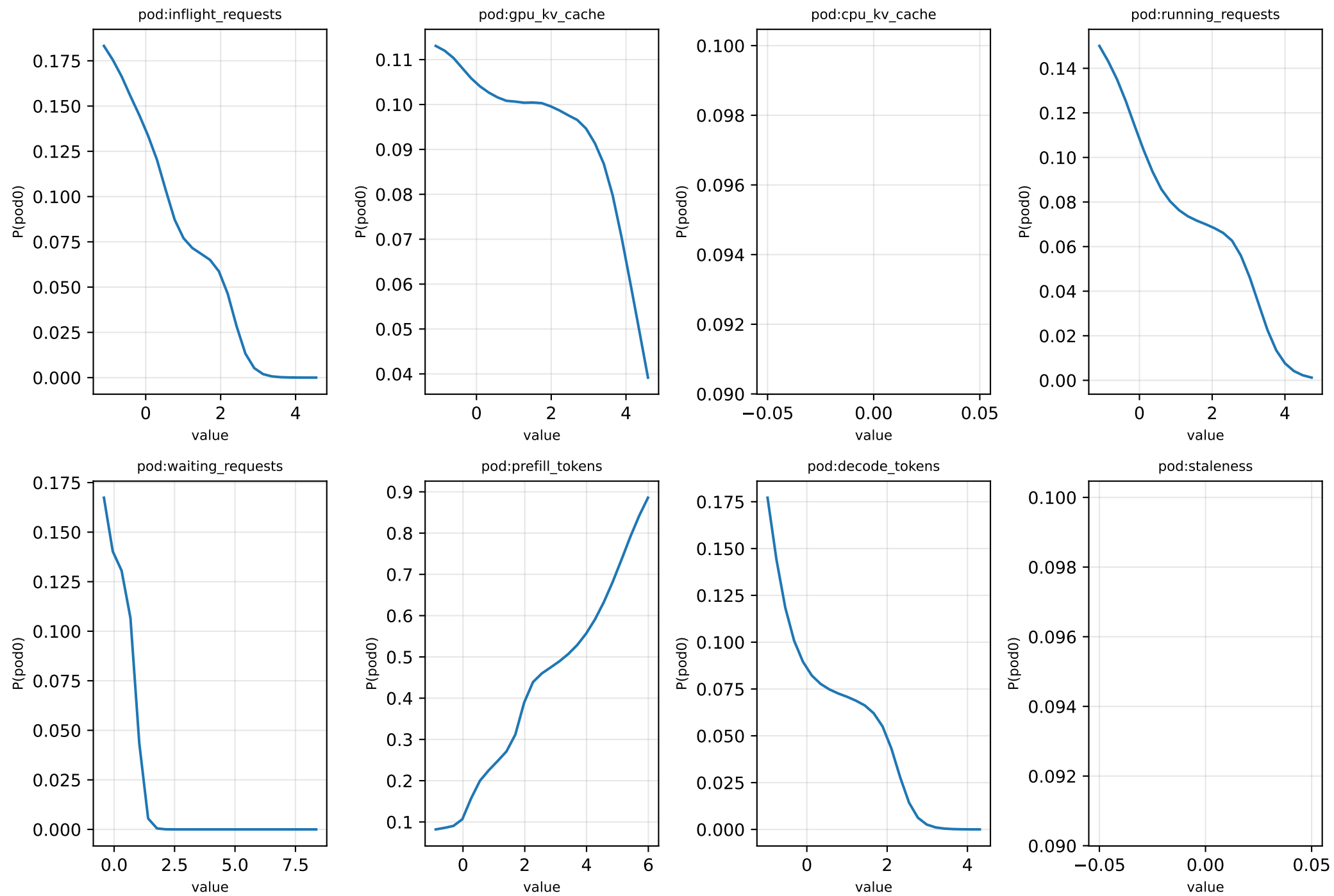
Permutation Symmetry Test

match_rate: 1.0

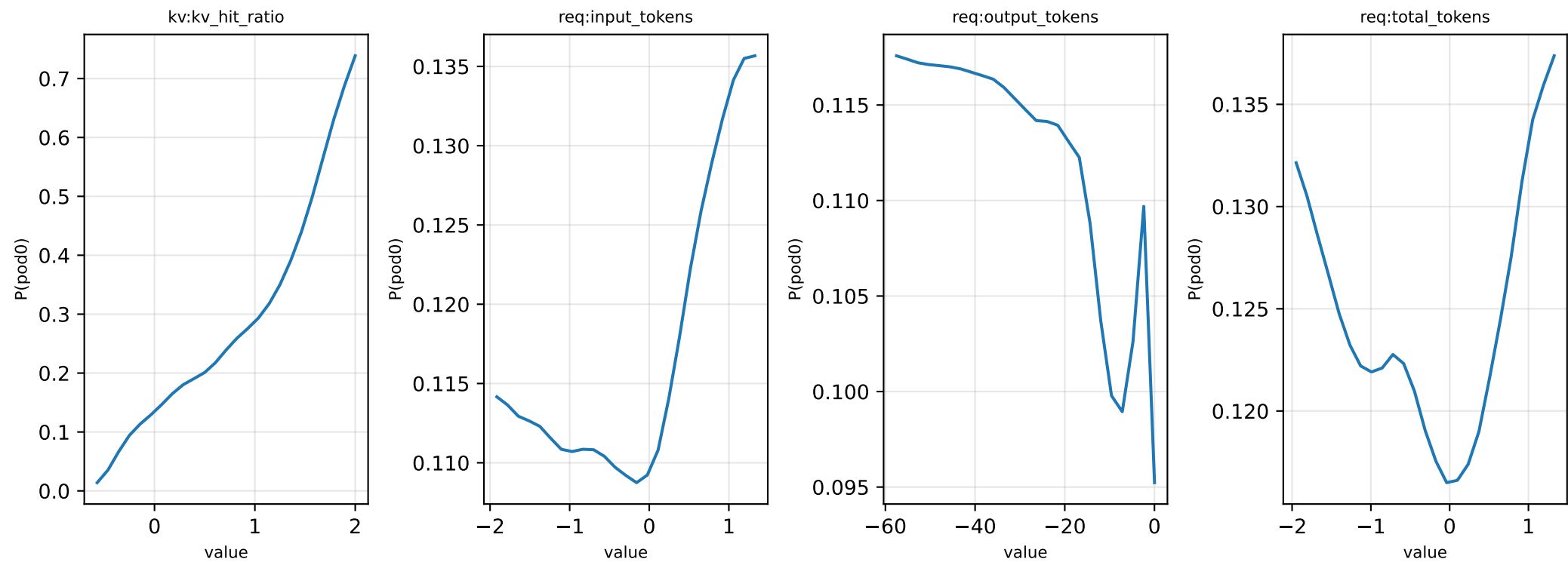
mean_l1_diff: 1.7054415957989748e-09

Higher match_rate and lower mean_l1_diff indicate invariance to pod ordering.

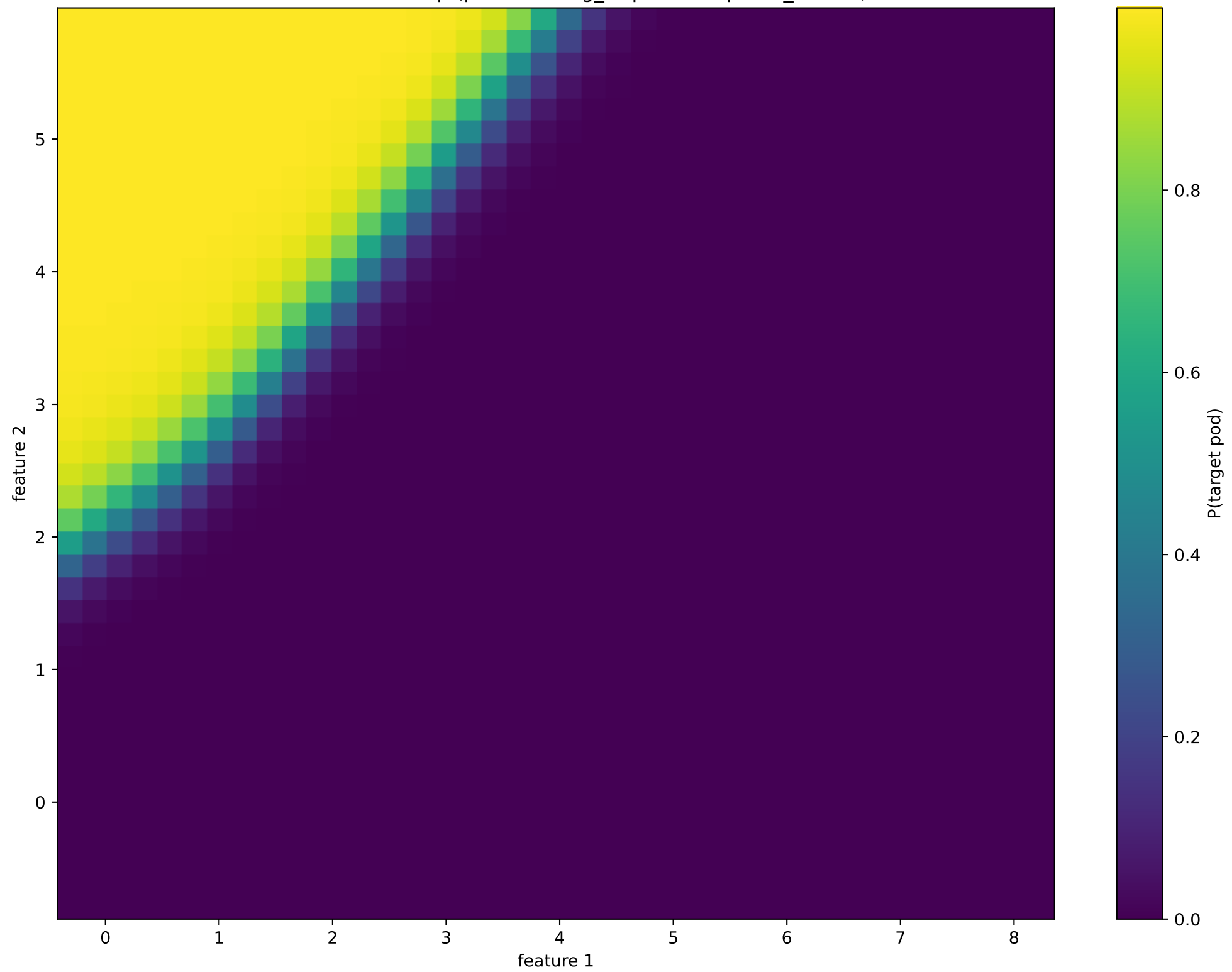
PDPs (all groups), target pod=0



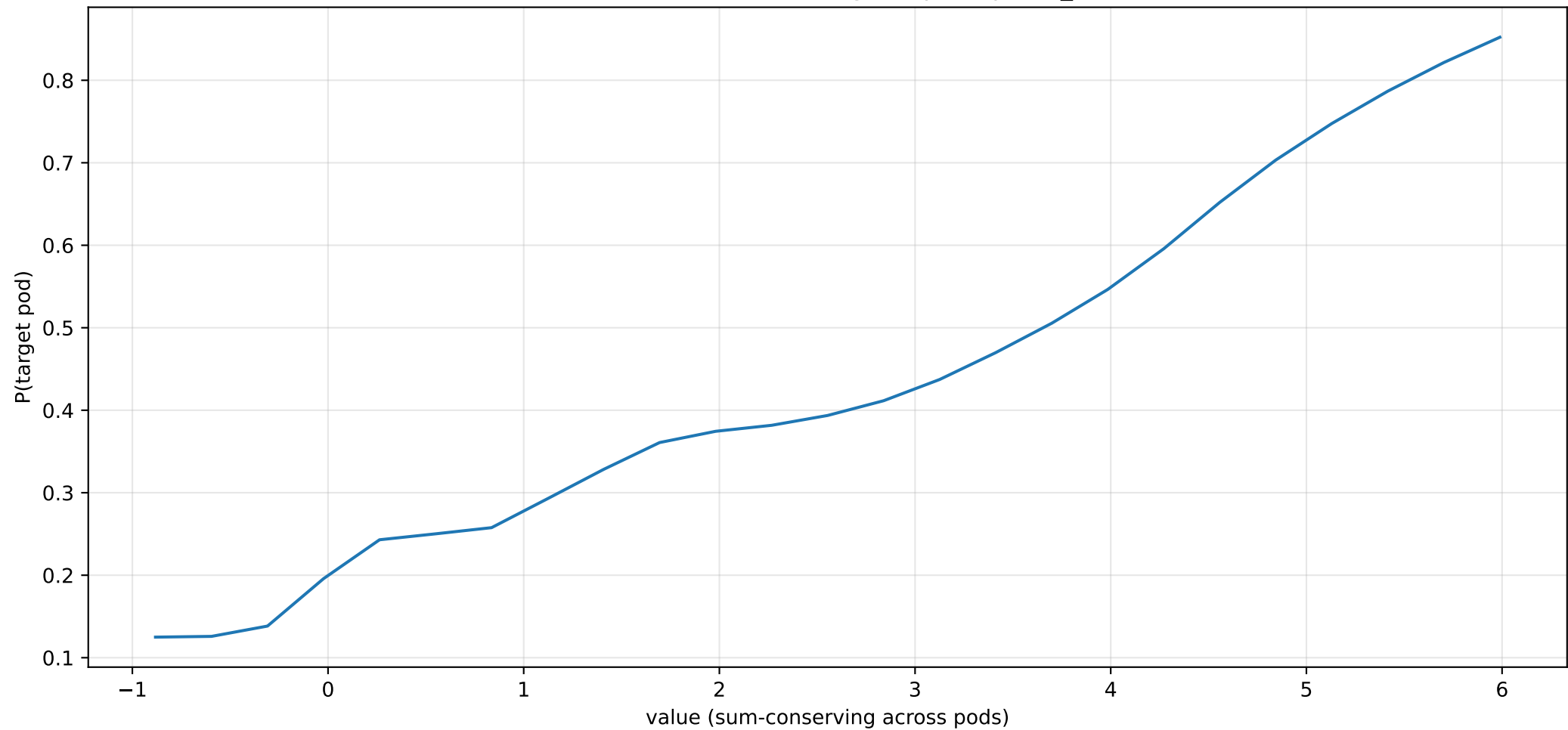
PDPs (all groups), target pod=0



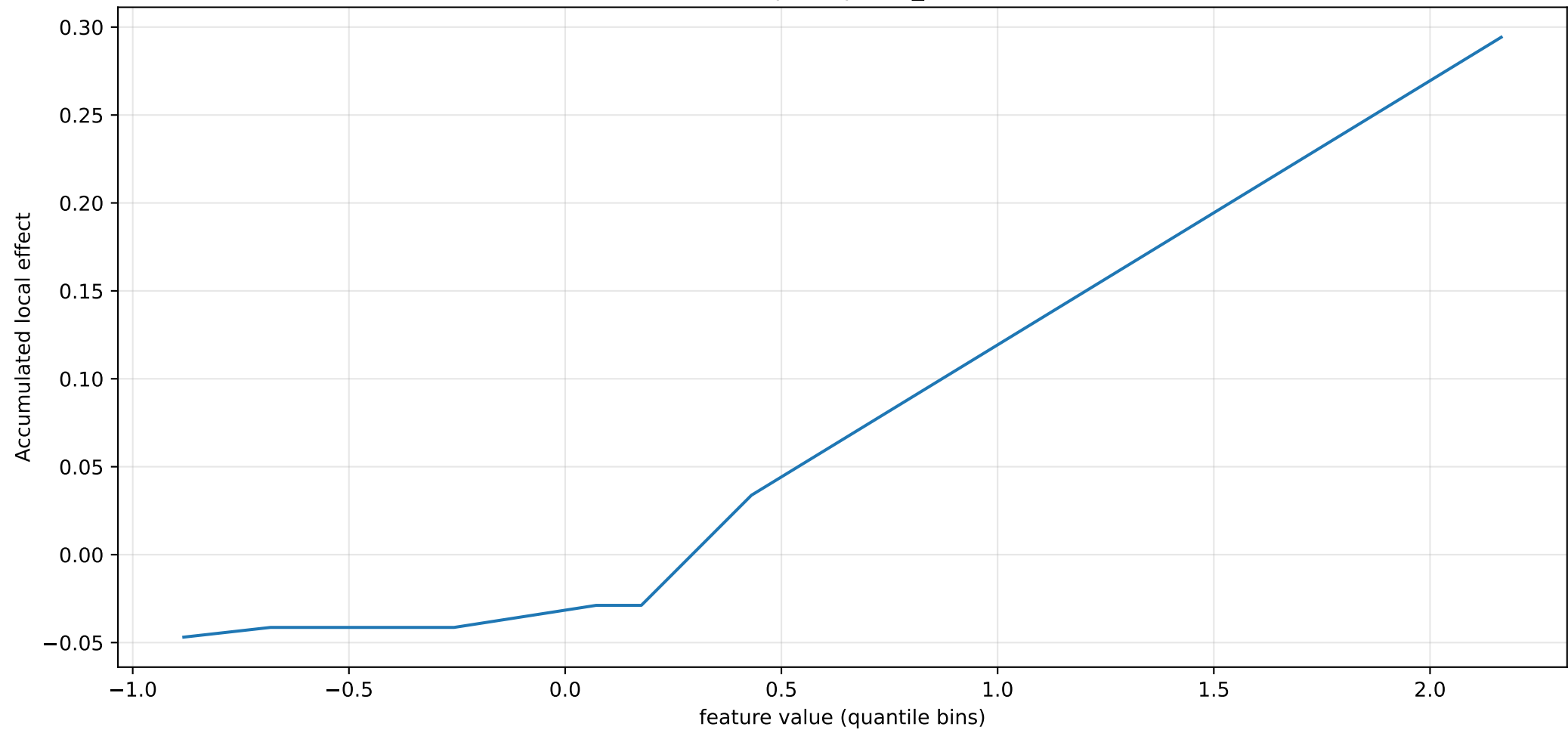
Interaction heatmap (pod0 waiting_requests vs prefill_tokens)



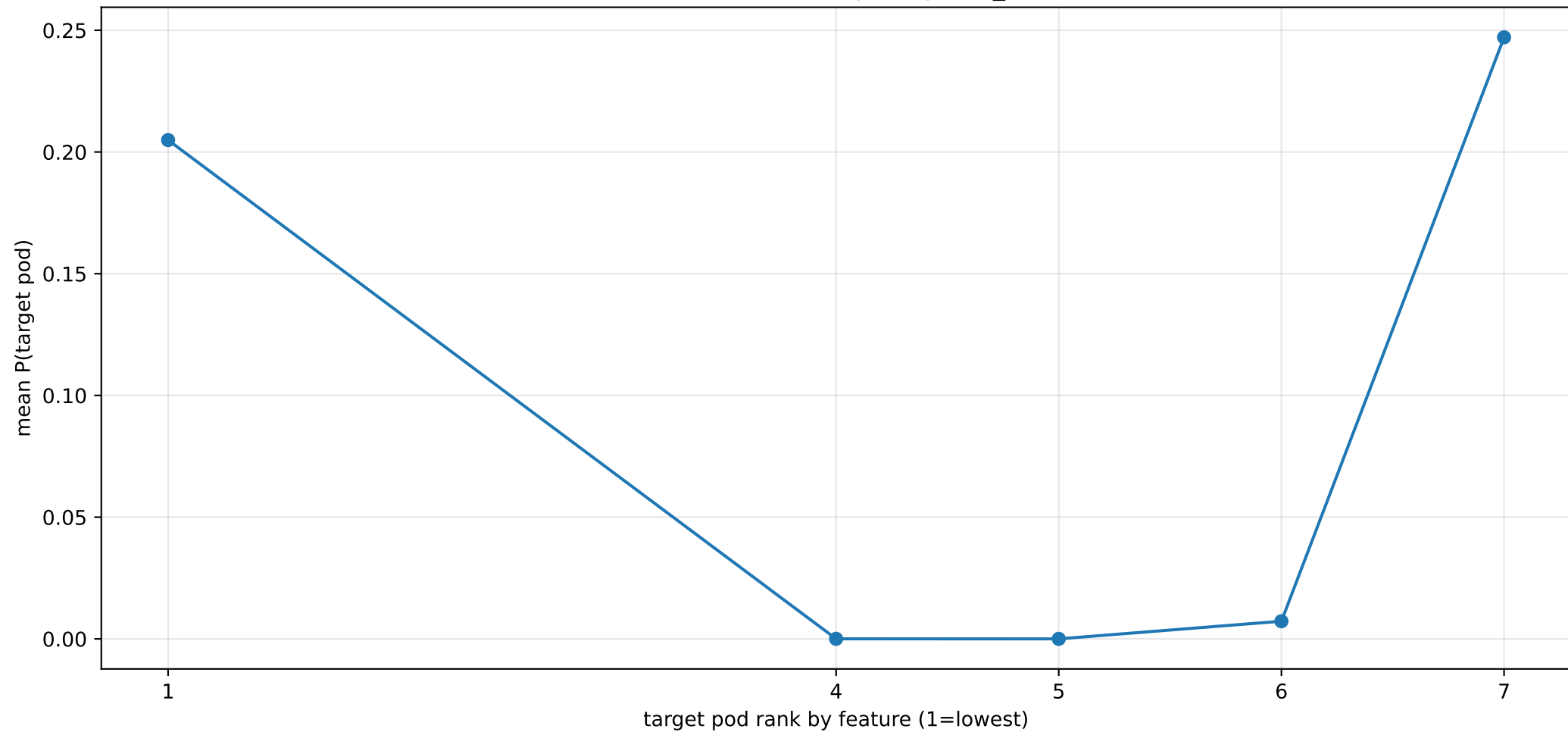
Relative PDP (sum-conserving) for pod0:prefill_tokens



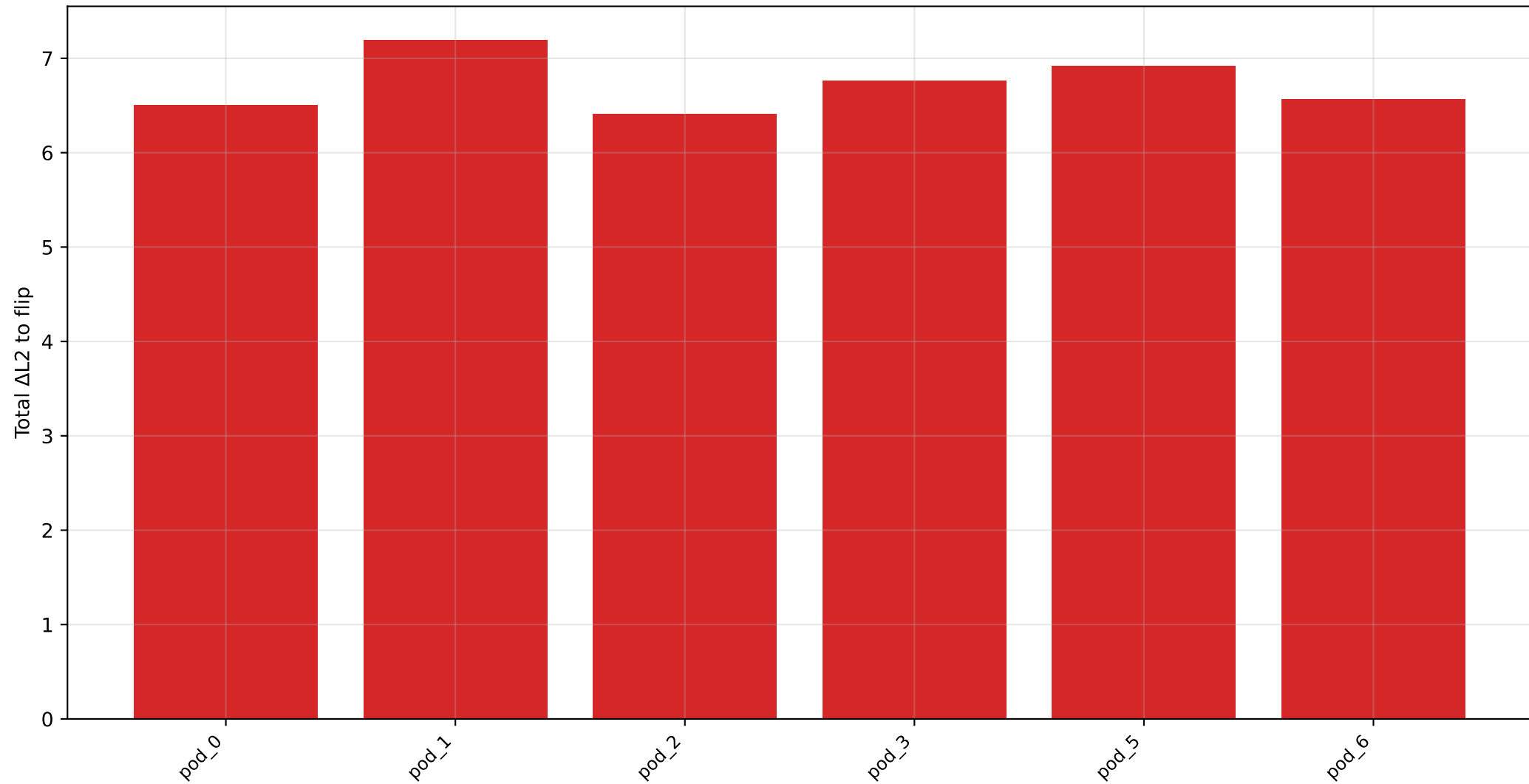
ALE for pod0:prefill_tokens



Rank-based effect for pod0:prefill_tokens



Counterfactual Distance to Flip to Each Pod (green=success, red=failed within budget)



EXPLAINABILITY FINDINGS

Saliency (local sensitivity):

- waiting_requests: 0.049
- prefill_tokens: 0.038
- decode_tokens: 0.031
- inflight_requests: 0.020
- running_requests: 0.013
- gpu_kv_cache: 0.008
- cpu_kv_cache: 0.002
- staleness: 0.001

Request Saliency:

- total_tokens: 0.007
- output_tokens: 0.005
- input_tokens: 0.005

KV Saliency:

- kv_hit_ratio: 0.096

Integrated Gradients (robust attribution):

- Top pod feature: waiting_requests
- KV importance: 0.392
- Request max attribution: output_tokens

PDP Trend Summary (target pod)

Pod features:

- inflight_requests: decreasing
- gpu_kv_cache: decreasing
- cpu_kv_cache: flat
- running_requests: decreasing
- waiting_requests: decreasing
- prefill_tokens: increasing
- decode_tokens: decreasing
- staleness: flat

KV features:

- kv_hit_ratio: increasing

Request features:

- input_tokens: increasing
- output_tokens: decreasing
- total_tokens: increasing

Permutation Symmetry:

- match_rate: 1.0
- mean_l1_diff: 1.7054415957989748e-09

Counterfactual Robustness:

- success_rate (flip achieved): 0.00
- min $\Delta L2$ to flip: 6.41
- mean $\Delta L2$ to flip: 6.72

Interpretation:

- High symmetry indicates no position bias.
- Larger $\Delta L2$ implies locally robust decisions (harder to flip).